SMART WASTE CLASSIFIER

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OUTLINE

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INTRODUCTION

- Waste management is a major challenge due to population growth and urbanization.
- Traditional methods cause inefficiencies like overflowing bins, poor segregation, and high costs.
- **Smart Waste Classifier** use sensors, IoT, and data analytics to improve collection, segregation, and disposal.
- These systems promote cleaner cities, reduce environmental impact, and support sustainability.

OBJECTIVES

- Waste detection with the help of a camera and image recognition.
- Automated sorting for food and plastic wastes.
- Setting up an alert mechanism to detect the presence of harmful gases using a gas sensor.
- Real-time monitoring.
- Tracking and detection using a GPS module.

LITERATURE SURVEY

- Waste Classification Using Deep Learning on Raspberry Pi
- Author(s): Jane Smith et al

➢ This paper explores how a Raspberry Pi with a camera module can be used to classify waste (plastic, food, metal, etc.) using deep learning models.

- Smart Garbage Bin for Efficient Waste Collection and Segregation
- Author(s): Rahul Kumar et al.

➢ This paper presents a smart garbage bin system that integrates ultrasonic sensors for waste level detection, gas sensors for odor monitoring, and AI-based waste classification.

LITERATURE SURVEY

- Smart Waste Management System Using IoT and Sensors
- Author(s): John Doe et al.

➢ This paper discusses an IoT-based smart waste management system that utilizes ultrasonic sensors to detect waste levels, gas sensors to monitor odor levels, and GPS for bin tracking

- Integration of GPS and IoT for Smart City Waste Management
- Author(s): Michael Brown et al.

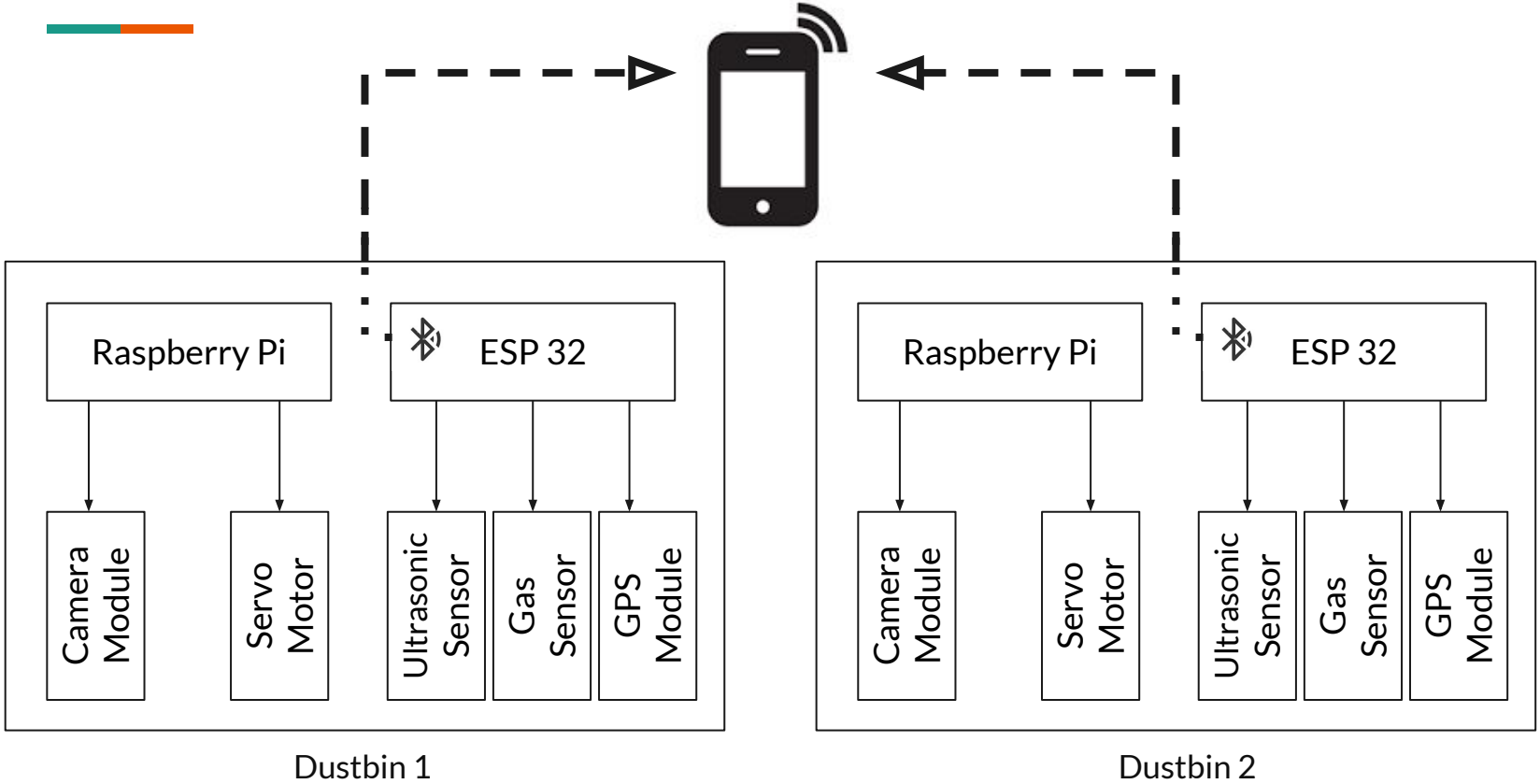
➢ This study investigates the role of GPS in smart waste management systems. It explains how GPS-enabled bins can track their locations and send alerts when full.

LITERATURE SURVEY

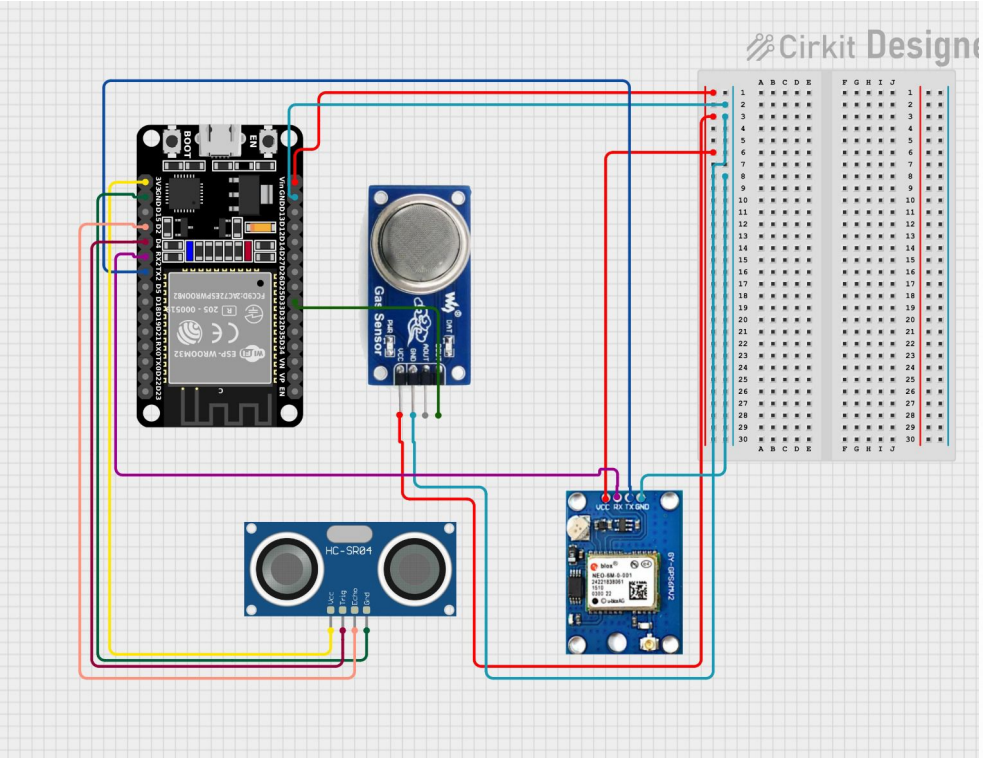
This project stands out by integrating multiple smart technologies—GPS, ultrasonic sensors, gas sensors, and AI-based waste classification—into a single system. This combination enhances waste management, making it more intelligent, efficient, and environmentally friendly.

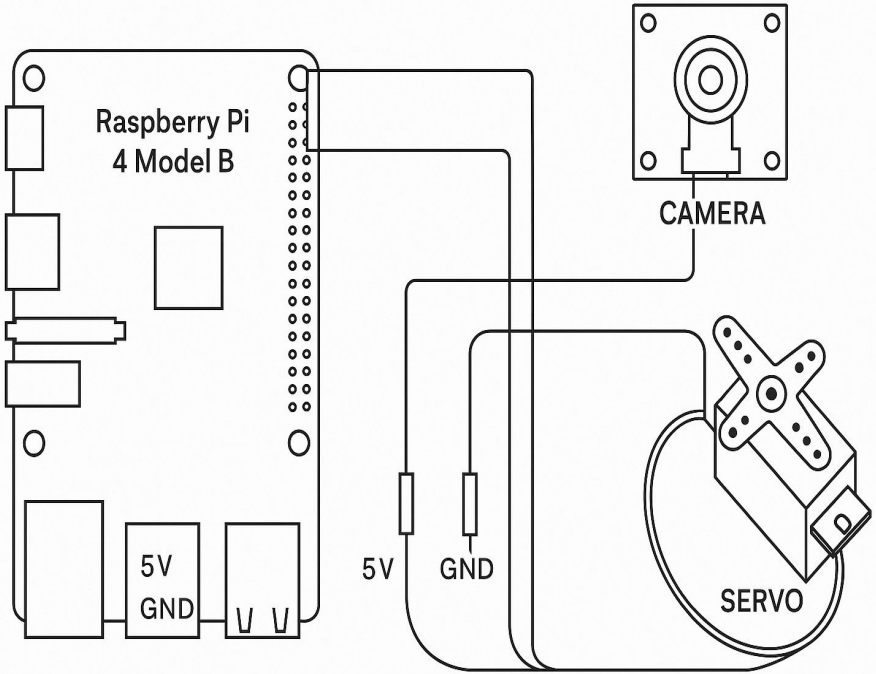
SYSTEM BLOCK DIAGRAM

SYSTEM BLOCK DIAGRAM



CIRCUIT DIAGRAM





Sensor Integration

1) HC-SR04 (Ultrasonic Sensor)

- **Formula for Distance Calculation:**
- **Threshold: 20 cm**
  - **< 20 cm → Too Close**
  - **≥ 20 cm → Safe Distance**

$$\text{Distance} = \frac{\text{Pulse Duration} \times 0.0343}{2}$$

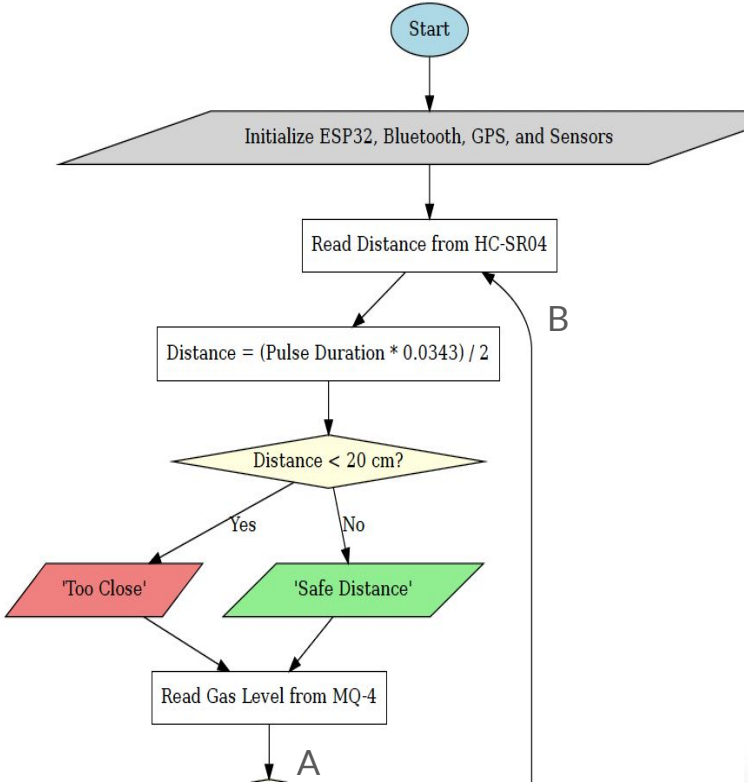
2) MQ-4 (Methane Gas Sensor)

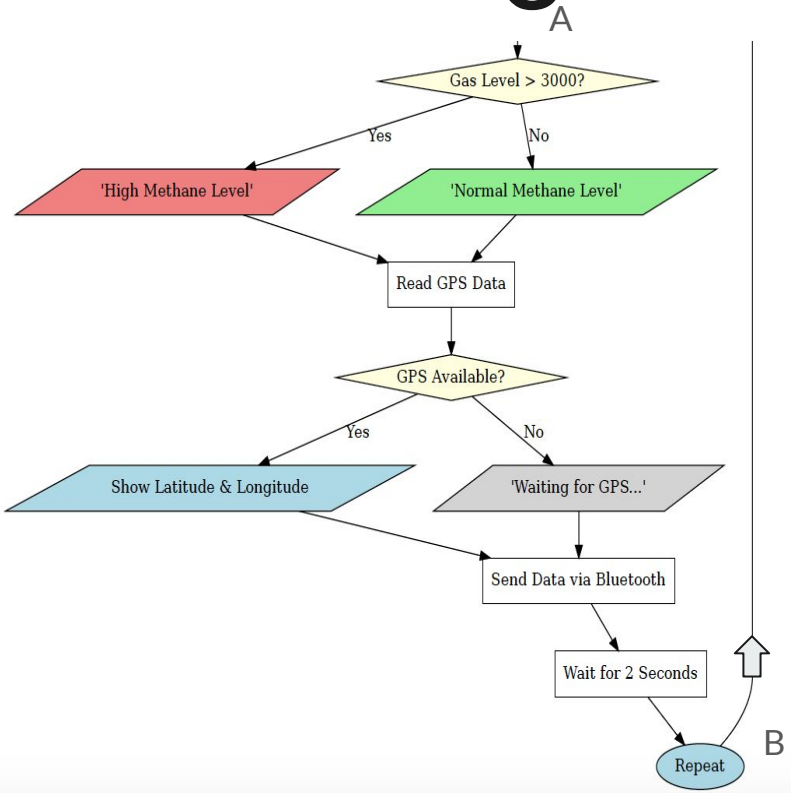
- **Reads Analog Gas Levels.**
- **Threshold: 3000**
  - **> 3000 → High Methane Level**
  - **≤ 3000 → Normal Methane Level**

3) GPS Module

- **Reads real-time GPS location.**
- **Outputs: Latitude & Longitude** → Sent via Bluetooth.

FLOWCHART - Sensor Integration





Waste Identification and Motor Integration

1. **Data Collection & Preprocessing**
   - Collect images of degradable and non-degradable waste (e.g., from Kaggle).
   - Resize images, normalize pixel values, and apply data augmentation.
   - Split dataset into training and testing sets.

2. **Model Training & Conversion**
   - Train a lightweight CNN for waste classification.
   - Convert the trained model to TensorFlow Lite (.tflite) for Raspberry Pi compatibility.

3. **Deploy Model on Raspberry Pi**
   - Transfer the trained **.tflite** model to Raspberry Pi.
   - Install necessary libraries (TensorFlow Lite, OpenCV).
   - Connect and test the Raspberry Pi camera module.

Waste Identification and Motor Integration

4. **Real-Time Waste Detection**
   - Capture an image from the camera.
   - Preprocess the image (resize, normalize) and feed it into the TensorFlow Lite model.
   - Perform classification to determine whether the waste is degradable or non-degradable.

5. **Servo Motor Control Based on Classification**
   - If the waste is degradable, trigger the servo motor to open the **degradable bin** lid.
   - If the waste is non-degradable, trigger the servo motor to open the **non-degradable** bin lid.

6. **Continuous Operation**
   - The system continuously captures images, classifies waste, and operates the correct bin lid until manually stopped.

Timeline of the Project

- **Implementation the ESP 32**
  Implemented the various sensors such as gas sensor, ultrasonic sensor and gps module into ESP32 board.(17/03/2025) [COMPLETED]
- **Implementation of Raspberry Pi**
  Addition of servo motors and camera module along with identification and classification of various waste materials using kaggle machine learning data set. (02/04/2025)
- **Final product Form**
  The final product form implementing both the ESP32 and Raspberry pi module along with soldered parts and functioning lids.(09/04/2025)
- **Report Submission**
  Document the system architecture, test results and future improvements in a structured report. (09/04/2025)

CONCLUSION

- **Efficient Waste Classification**: Successfully implemented real-time waste detection using **TensorFlow Lite and OpenCV** on **Raspberry Pi 4 Model B**.
- **Automated Bin Operation**: Servo-controlled smart bins ensure proper waste segregation into **degradable** and **non-degradable** categories.
- **Integrated Multi-Sensor System**: ESP32 manages **ultrasonic, gas (MQ-4), and GPS sensors**, enhancing monitoring and safety.
- **IoT & Wireless Connectivity**: Real-time data visualization via **Bluetooth output** for remote access.
- **Portable & Standalone System**: Fully functional without external computing devices, making it **cost-effective and scalable**.